

Ethical Decision-Making Choices in Engineering

Introduction

Ethical decision-making means choosing the most morally correct action when an engineer faces a difficult situation. In engineering, problems are often not simple because an engineer may have duties toward the employer, client, public, law, profession, and environment at the same time.

According to Fleddermann, engineering ethics helps future engineers face and resolve ethical dilemmas in professional life. Ethical problems are also like engineering design problems because there is often no single perfect answer; instead, there may be several possible choices, some clearly right, some better, and some clearly wrong.

1. Need for Ethical Decision-Making

Engineers make decisions that can affect public safety, health, business, environment, and society. Sometimes the correct action is obvious, such as refusing to falsify test results. But many engineering cases involve conflict between two duties.

For example, an engineer may want to remain loyal to the employer, but the same engineer may also have a duty to protect the public. In such cases, ethical decision-making helps the engineer choose the action that best follows professional responsibility.

Fleddermann says the purpose of studying engineering ethics is to develop **moral autonomy**, which means the ability to think critically and independently about moral issues in professional practice.

2. Step 1: Identify the Ethical Problem

The first step is to clearly understand the problem. An engineer should not directly jump to a decision without knowing what is actually wrong.

The engineer should ask:

- What happened?
- Who is affected?
- What duties are involved?
- Is anyone being harmed?

- Is there pressure from employer or client?
- Is the action legal but still unethical?

This step is important because a wrong understanding of the problem can lead to a wrong ethical decision.

3. Step 2: Analyze the Issues

Fleddermann explains that ethical problems can be analyzed through **three types of issues**:

A. Factual Issues

Factual issues deal with what is actually true. The engineer must collect facts, data, reports, test results, laws, and technical evidence.

For example, before declaring a product unsafe, the engineer should check whether the test data actually proves a safety risk.

B. Conceptual Issues

Conceptual issues deal with the meaning of words or ideas. Sometimes the problem depends on how we define a concept.

For example, is a lunch from a supplier only a gift, or is it a bribe? Is certain information public, or is it confidential?

C. Moral Issues

Moral issues deal with which ethical principle applies. After facts and concepts are clear, the engineer must decide what is right or wrong.

For example, if the “gift” is actually a bribe, then the moral decision is clear: it should not be accepted.

4. Step 3: Use Codes of Ethics

Codes of ethics provide a framework for ethical judgment. They explain the duties and responsibilities of engineers, especially toward public safety, honesty, competence, fairness, and professional conduct.

However, Fleddermann clearly explains that a code of ethics is **not a complete recipe** for every situation. It is only a guide and starting point. Engineers must still use sound judgment because no code can cover every possible ethical problem.

5. Step 4: Consider the Possible Choices

After understanding the problem, the engineer should list all possible actions. Ethical decision-making is not only about choosing between “do it” or “do not do it.” Sometimes there may be a better middle solution.

Possible choices may include:

- accept the decision
- refuse the decision
- ask for more testing
- inform the manager
- consult senior engineers
- follow company policy
- report to higher authority
- withdraw from the project
- whistleblow in serious cases

The best choice should protect public safety, follow professional duties, and avoid dishonesty or harm.

6. Step 5: Use Ethical Problem-Solving Techniques

Fleddermann gives useful methods for solving ethical problems.

A. Line Drawing Technique

Line drawing is used when the situation is unclear. In this method, the engineer compares the case with a clearly ethical action and a clearly unethical action.

One side is the **positive paradigm**, meaning the clearly right action. The other side is the **negative paradigm**, meaning the clearly wrong action. Then the actual case is placed between them to judge where it lies.

This method is useful, but it must be used honestly and objectively. Fleddermann warns that dishonest use of line drawing can make a wrong action appear right.

B. Flow Charting

Flow charting is useful when a decision has many steps and consequences. It helps the engineer see what may happen after each choice.

For example, if a safety system is removed for maintenance, the engineer can make a flow chart showing whether other safety systems are working, whether danger exists, and whether maintenance should continue or stop.

Fleddermann explains that flow charts are helpful because they visually show the consequences of each decision.

C. Conflict Problem Method

Sometimes two moral duties conflict with each other. Fleddermann explains that conflict problems can be solved in different ways.

First, if one duty is clearly more important, choose the stronger duty. For example, public safety is more important than loyalty to employer.

Second, the engineer can find a **creative middle way**, meaning a compromise that protects the main moral values of both sides.

Third, if no compromise is possible, the engineer must choose the action that best protects the most important ethical duty.

7. Step 6: Make and Defend the Final Decision

After analyzing facts, concepts, moral principles, codes, and consequences, the engineer should choose the most ethical option.

A good final decision should be:

- safe for the public
- honest and truthful
- legally acceptable
- professionally responsible
- fair to all affected people
- defensible if questioned later

A useful test is: **Can I defend this decision in front of my teacher, employer, court, professional body, or the public?**

How to Apply in Case Studies

In exam case studies, use this structure:

First, write the main ethical conflict. For example, “The conflict is between company profit and public safety.”

Second, identify stakeholders such as engineer, employer, client, public, users, government, and environment.

Third, mention factual, conceptual, and moral issues. Explain what facts are needed, what terms must be defined, and what moral duty applies.

Fourth, apply codes of ethics and ethical theories. Usually, public safety, honesty, and professional responsibility are the strongest points.

Finally, give a clear decision. Do not write a confused answer. Choose one side and justify it ethically.

Short Case Study: Ralph and SDX Alliance

Ralph reused software code from his previous company in the SDX Alliance product. Later, it was found that the previous company owned the copyright.

The factual issue is that the code belongs to the previous company. The conceptual issue is whether Ralph had the right to reuse code that he personally wrote. The moral issue is whether using someone else’s intellectual property without permission is ethical.

SDX Alliance should not release the product using that code without permission. The company should try to contact the copyright owner, get legal permission, or rewrite the code. Even if the delay is costly, releasing copyrighted code would violate honesty, professional responsibility, and respect for intellectual property.

Memorization Checklist

Remember ethical decision-making with:

F-C-M-C-C-D

F — Facts: Find what actually happened.

C — Concepts: Define unclear terms like gift, bribe, safety, confidentiality.

M — Moral issue: Decide which ethical duty applies.

C — Codes: Apply professional codes of ethics.

C — Choices: List possible actions and consequences.

D — Decision: Choose and defend the most ethical action.

5-Mark Exam Answer

Ethical decision-making means choosing the morally correct action when an engineer faces a professional dilemma. According to Fleddermann, engineering ethics prepares engineers to analyze and resolve ethical problems in professional life. These problems are like design problems because they may not have one perfect answer; instead, engineers must choose the best option among several possible choices.

The first step is to identify the problem and stakeholders. Then the engineer should analyze factual, conceptual, and moral issues. Factual issues deal with what is true, conceptual issues deal with the meaning of terms, and moral issues deal with the ethical principle that applies. After this, the engineer should use codes of ethics, but codes are only a framework and not a substitute for sound judgment.

Engineers can also use line drawing, flow charting, and conflict problem methods. Line drawing compares the case with clearly right and clearly wrong actions. Flow charting shows the consequences of each choice. Conflict problem solving helps when two duties clash, such as loyalty to employer and public safety. In the end, the engineer should choose the action that protects public welfare, honesty, fairness, and professional responsibility.